

osTicket Help Desk Lab

Setup, Configuration & Workflow Documentation

Prepared by: **Bryan Sykes**

Environment: Local XAMPP | osTicket v1.18.x | Windows

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PROJECT OBJECTIVE: Deploy, configure, and demonstrate a fully functional IT help desk ticketing system using osTicket on a local XAMPP stack. This project simulates a real-world enterprise helpdesk environment including departments, agents, SLA plans, automation rules, and end-to-end ticket workflows.

Technology Stack

Component	Details
Web Server	Apache (via XAMPP 8.1.x)
Database	MySQL 8.x (via XAMPP)
Language Runtime	PHP 8.1
Ticketing Platform	osTicket v1.18.x
DB Management	phpMyAdmin
Host OS	Windows 10/11
Deployment Type	Local (localhost)

Section 1: Installation & Setup

1.1 Environment Preparation

osTicket is a web application, not a traditional Windows program. It requires a web server (Apache), PHP runtime, and a MySQL database to operate. XAMPP bundles all three into a single Windows installer, making it the fastest path to a working local deployment.

XAMPP Installation Steps

- Downloaded XAMPP 8.1.x from apachefriends.org
- Ran the installer with default settings, installed to C:\xampp
- Launched XAMPP Control Panel and started Apache and MySQL services
- Confirmed both services showed green status indicators

1.2 PHP Extension Configuration

osTicket requires several PHP extensions disabled by default in XAMPP. The following were enabled by editing php.ini (XAMPP Control Panel → Apache Config → PHP (php.ini)):

```
extension=imap
extension=mysqli
extension=gd
extension=gettext
extension=intl
extension=mbstring
```

After saving php.ini, Apache was restarted via the XAMPP Control Panel to reload the configuration.

1.3 Database Setup

- Navigated to <http://localhost/phpmyadmin>
- Created a new database named: osticket
- Set collation to: utf8mb4_unicode_ci

1.4 Post-Installation Cleanup

- Deleted the setup\ folder: C:\xampp\htdocs\osticket\setup\
 - Prevents anyone from re-running the installer and overwriting the configuration
- Set ost-config.php to read-only via Windows file properties
 - Prevents accidental or unauthorized modification of core configuration

Section 2: Troubleshooting Log

The following issues were encountered and independently resolved during the installation process. Each represents a real-world diagnostic scenario demonstrating systematic problem-solving.

Issue 1 — 404 Error on /osticket/setup/

Root Cause: osTicket was installed by copying the upload\ and scripts\ parent folders into htdocs\osticket\ rather than copying the contents of upload\ directly into htdocs\osticket\.

Incorrect Directory Structure

```
C:\xampp\htdocs\osticket\  
  upload\          <- Wrong: folder copied, not its contents  
    index.php  
    bootstrap.php  
    include\  
      setup\  
scripts\  

```

Correct Directory Structure

```
C:\xampp\htdocs\osticket\  
  index.php        <- Correct: contents of upload\ pasted directly here  
  bootstrap.php  
  include\  
  setup\  
scripts\          <- scripts\ stays at this level intentionally  

```

Resolution

- Opened C:\xampp\htdocs\osticket\upload\
- Selected all contents (Ctrl+A) and cut (Ctrl+X)
- Pasted directly into C:\xampp\htdocs\osticket\
- Deleted the now-empty upload\ folder
- Confirmed <http://localhost/osticket/setup/> loaded the web installer

Key learning: osTicket's web root is the contents of the upload folder, not the folder itself. This reflects standard web application deployment practice where the application root must be served directly by the web server.

Issue 2 — HTTP 500 / PHP Fatal Error (Wrong Database Hostname)

After correcting the directory structure, the installer returned an HTTP 500 Internal Server Error. The Apache error log was checked via XAMPP Control Panel → Logs → Apache Error Log:

```
[Mon Apr 27 11:44:03.720288 2026] [mpm_winnt:notice] [pid 71196:tid 492] AH00418: Parent: Created child process 21292  
[Mon Apr 27 11:44:03.949306 2026] [ssl:warn] [pid 21292:tid 504] AH01909: www.example.com:443:0 server certificate does NOT include an ID  
which matches the server name  
[Mon Apr 27 11:44:03.972306 2026] [ssl:warn] [pid 21292:tid 504] AH01909: www.example.com:443:0 server certificate does NOT include an ID  
which matches the server name  
[Mon Apr 27 11:44:03.996648 2026] [mpm_winnt:notice] [pid 21292:tid 504] AH00354: Child: Starting 150 worker threads.  
[Mon Apr 27 11:45:21.838378 2026] [php:error] [pid 21292:tid 1980] [client ::1:51009] PHP Fatal error: Uncaught mysqli_sql_exception:  
php_network_getaddresses: getaddrinfo for port80 failed: No such host is known. in C:\xampp\htdocs\osticket\include\mysqli.php:69  
\nStack trace:\n#0 C:\xampp\htdocs\osticket\include\mysqli.php(69): mysqli->real_connect('port80', 'root',  
Object(SensitiveParameterValue), NULL, '3306', '')\n#1 C:\xampp\htdocs\osticket\setup\inc\class.installer.php(94):  
db_connect('port80', 'root', 'root')\n#2 C:\xampp\htdocs\osticket\setup\install.php(52): Installer->install(Array)\n#3 {main}\nthrown in C:\xampp\htdocs\osticket\include\mysqli.php on line 69, referer: http://localhost/osticket/setup/
```

Root Cause: The Database Host field in the osTicket installer was entered as 'port80' instead of 'localhost'. PHP attempted to resolve 'port80' as a server hostname, which does not exist on the local network.

Resolution

- Returned to <http://localhost/osticket/setup/>
- Corrected the Database Host field to: localhost
- Verified remaining database fields were correct before resubmitting

Issue 3 — Installer Rejected Blank Database Password

Root Cause: Newer versions of osTicket enforce a non-blank MySQL password as a security requirement. The installer will not proceed with an empty password field.

Resolution

- Opened XAMPP Shell from the Control Panel
- Set a MySQL root password using the mysqladmin CLI tool:

```
mysqladmin -u root password "osticket123"
```

- Updated phpMyAdmin config to match the new password:

```
// File: C:\xampp\phpMyAdmin\config.inc.php
$config['Servers'][$i]['password'] = 'osticket123';
```

Issue 4 — phpMyAdmin Access Denied After Password Change

After setting the MySQL root password, phpMyAdmin returned an access denied error because its configuration file had not yet been updated to reflect the new credential:



Root Cause: phpMyAdmin's config.inc.php still referenced a blank password while MySQL now required 'osticket123'. The credential mismatch caused authentication failure on every connection attempt.

Resolution

- Opened C:\xampp\phpMyAdmin\config.inc.php in a text editor
- Updated the password field to match the MySQL root password set in Issue 3
- Saved the file and refreshed <http://localhost/phpmyadmin>
- phpMyAdmin access restored and database confirmed accessible

Production Note: In a production environment, MySQL root would never be directly exposed to phpMyAdmin with a simple password. A dedicated application database user with scoped permissions would be created instead. This simplified configuration is appropriate for a local lab only.

Section 3: System Configuration

3.1 Departments

Departments represent the functional teams that own and handle tickets. Four departments were created to mirror a realistic IT organizational structure:

Department	Purpose
IT Support	Tier 1 general helpdesk — first point of contact for all users
Systems Administration	Tier 2 escalations — server, workstation, and systems issues
Network Operations	Network and connectivity issues
Security	Security incidents, access management, and threat response

3.2 Teams

Teams operate across departments and are used for ticket routing, escalation, and specialized groupings:

Team	Function
Tier 1 Support	First-line response — general troubleshooting and standard requests
Tier 2 Support	Escalated technical issues across sysadmin and network departments
Incident Response	Security incidents and critical outages requiring urgent response

3.3 Agents

Four agents were created representing a realistic helpdesk team, each mapped to a specific department and team:

Agent	Department Team
Alex Turner	IT Support Tier 1 Support
Lindsey Matthews	Systems Administration Tier 2 Support
Sam Rivera	Network Operations Tier 2 Support
Sarah Barker	Security Incident Response

3.4 Users (End Users)

Four users were created to simulate employees submitting helpdesk tickets:

User	Email
James Carter	james.carter@company.com
Derek Walsh	derek.walsh@company.com
Kayla Jefferies	kayla.jefferies@company.com
Yvette Cheatham	yvette.cheatham@company.com

3.5 SLA Plans

Three SLA tiers were configured to enforce response and resolution time requirements. The SLA engine automatically calculates due dates when tickets are created based on the assigned plan:

SLA Plan	Grace Period Schedule Use Case
SEV-A (Critical)	1 hour 24/7 Critical outages, security incidents, system-wide failures
SEV-B (High)	4 hours 24/7 Significant issues affecting multiple users or key systems
SEV-C (Normal)	8 hours Mon–Fri 8am–5pm Standard requests, minor issues, single user problems

3.6 Help Topics

Six help topics were created and mapped to departments, SLA plans, and priority levels. Users select a help topic when submitting tickets, which triggers the corresponding automation filter:

Help Topic	Department SLA Priority
Business Critical Outage	IT Support SEV-A Emergency
Security Incident	Security SEV-A Emergency
Password Reset	IT Support SEV-B High
Network Connectivity Issue	Network Operations SEV-B High
Personal Computer Issues	IT Support SEV-C Normal
Equipment Request	IT Support SEV-C Low

Section 4: Automation Rules

Four ticket filters were configured to automatically route incoming tickets based on the help topic selected at submission. Automation handles department assignment, team assignment, priority, and SLA plan — eliminating manual triage for common ticket types and ensuring consistent, error-free routing.

Filter 1 — Security Incident Auto-Route

Setting	Value
Filter Name	Security Incident Auto-Route
Execution Order	1
Target Channel	osTicket Alerts
Criteria	Help Topic is Security Incident
Actions	Department → Security Team → Incident Response Priority → Emergency SLA → SEV-A

Filter 2 — Network Issue Auto-Route

Setting	Value
Filter Name	Network Issue Auto-Route
Execution Order	2
Target Channel	osTicket Alerts
Criteria	Help Topic is Network Connectivity Issue
Actions	Department → Network Operations Team → Tier 2 Support Priority → High SLA → SEV-B

Filter 3 — Password Reset Auto-Assign

Setting	Value
Filter Name	Password Reset Auto-Assign
Execution Order	3
Target Channel	osTicket Alerts
Criteria	Help Topic is Password Reset
Actions	Department → IT Support Team → Tier 1 Support Priority → High SLA → SEV-B

Filter 4 — Critical Outage Escalation

Setting	Value
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Filter Name	Critical Outage Escalation
Execution Order	4
Target Channel	osTicket Alerts
Criteria	Help Topic is Business Critical Outage
Actions	Department → IT Support Team → Tier 2 Support Priority → Emergency SLA → SEV-A

Section 5: Ticket Workflow Simulations

Four end-to-end ticket scenarios were executed to validate the complete system configuration including user submission, automated routing, agent assignment, internal documentation, and user-facing communication.

Scenario 1 — Password Reset (Tier 1 | SEV-B | High)

Submitted by: Kayla Jefferies

Field	Value
Help Topic	Password Reset
Summary	Unable to log into my workstation
Auto-Routed To	IT Support Tier 1 Support
Priority	High
SLA Plan	SEV-B
Assigned Agent	Alex Turner

Full Ticket View:

 Ticket #790969



Unable to log into my workstation

Status:	Open	User:	 Kayla Jefferies (1)  (Manage Collaborators)
Priority:	High	Email:	kayla.jefferies@helpdesk.com
Department:	IT Support	Source:	Web (:::1)
Create Date:	4/27/26 12:54 PM		

Assigned To:	Alex Turner	Help Topic:	Password Reset
SLA Plan:	SEV-B (High)	Last Message:	4/27/26 12:54 PM
Due Date:	4/27/26 4:54 PM	Last Response:	4/27/26 1:02 PM

Ticket Thread (3)

Tasks

Kayla Jefferies posted 4/27/26 12:54 PM

I have been locked out of my workstation since this morning and cannot access any of my files or applications. I need my password reset as soon as possible.

 Created by  Kayla Jefferies 4/27/26 12:54 PM

  Bryan Sykes assigned this to Alex Turner 4/27/26 1:00 PM

Bryan Sykes posted 4/27/26 1:01 PM

I have been locked out of my workstation since this morning and cannot access any of my files or applications. I need my password reset as soon as possible.

Bryan Sykes posted 4/27/26 1:02 PM

Hi Kayla, this is Alex from IT Support. I've received your request and will be assisting you with your password reset today. Please expect a call within the hour.

Validation: Automation filter fired correctly on submission. Priority, department, and team all populated automatically. Agent assignment was performed manually by the queue manager — this reflects standard helpdesk practice where automation handles routing while a human agent claims or is assigned individual tickets from the queue.

Email Behavior Note: In this local deployment, agent replies are sent from the admin email address rather than individual agent addresses. In a production environment with a configured SMTP mail server, replies would originate from each agent's individual email identity. This is expected behavior in a lab without mail server configuration.

Scenario 2 — Security Incident (SEV-A | Emergency)

Submitted by: Derek Walsh

Field	Value
Help Topic	Security Incident
Summary	Suspicious login attempts on my account
Auto-Routed To	Security Incident Response
Priority	Emergency
SLA Plan	SEV-A
Assigned Agent	Sarah Barker

Ticket thread and audit trail:

Derek Walsh posted 4/27/26 1:05 PM

Hello there. I received multiple alerts about failed login attempts on my account from an unrecognized location. I did not initiate these and I am concerned my credentials may be compromised. I need some help as soon as possible.

Created by Derek Walsh 4/27/26 1:05 PM

Bryan Sykes assigned this to Sarah Barker 4/27/26 1:06 PM

Bryan Sykes posted 4/27/26 1:09 PM

Good Afternoon Derek, this is Sarah from the Security department. I've received your request and will be looking into those failed login attempts as soon as possible. please expect a call within the hour.

Validation: Ticket routed directly to Security department and Incident Response team. Priority escalated to Emergency automatically. Audit trail logged the assignment action with full timestamp. Sarah Barker responded acknowledging receipt and initiating investigation.

Scenario 3 — Network Connectivity Issue (SEV-B | High)

Submitted by: Yvette Cheatham

Field	Value
Help Topic	Network Connectivity Issue
Summary	Unable to connect to company network
Auto-Routed To	Network Operations Tier 2 Support
Priority	High
SLA Plan	SEV-B
Due Date	Auto-calculated: 4 hours from ticket creation
Assigned Agent	Sam Rivera

Full ticket view:

Ticket #674805

Unable to connect to company network

Status: Open

Priority: High

Department: Network Operations

Create Date: 4/27/26 1:13 PM

User: Yvette Cheatham (1) (Manage Collaborators)

Email: yvette.cheatham@helpdesk.com

Source: Web (::1)

Assigned To: Sam Rivera

SLA Plan: SEV-B (High)

Due Date: 4/27/26 5:13 PM

Help Topic: Network Connectivity Issue

Last Message: 4/27/26 1:13 PM

Last Response: 4/27/26 1:19 PM

Ticket Thread (2)

Tasks

Yvette Cheatham posted 4/27/26 1:13 PM

Good Afternoon,

Since this morning I have been unable to connect to the company network from my workstation. I cannot access any shared drives or internal applications. I have already tried restarting my machine with no success. I could really use some help as I have big project to submit by the end of the week.

Created by Yvette Cheatham 4/27/26 1:13 PM

Bryan Sykes assigned this to Sam Rivera 4/27/26 1:15 PM

Bryan Sykes posted 4/27/26 1:19 PM

Good Afternoon Yvette,

This is Sam Rivera from Network Operations. I have received your ticket and will be contacting you by phone shortly to help with any issues you may be dealing with.

Ticket is assigned to Sam Rivera

Validation: Ticket routed to Network Operations with Tier 2 Support team. The SLA engine auto-calculated a due date exactly 4 hours from submission, confirming SEV-B grace period is functioning correctly. Sam Rivera acknowledged the ticket and initiated user contact.

Scenario 4 — Business Critical Outage (SEV-A | Emergency)

Submitted by: James Carter

Field	Value
Help Topic	Business Critical Outage
Summary	Company-wide email system is completely down
Auto-Routed To	IT Support Tier 2 Support
Priority	Emergency
SLA Plan	SEV-A
Due Date	Auto-calculated: 1 hour from ticket creation
Assigned Agent	Lindsey Matthews

Full ticket view:

Ticket #442727

Company-wide email system is completely down

Status: Open

Priority: Emergency

Department: IT Support

Create Date: 4/27/26 1:21 PM

User: James Carter (1) (Manage Collaborators)

Email: james.carter@helpdesk.com

Source: Web (::1)

Assigned To: Lindsey Matthews

SLA Plan: SEV-A (Critical)

Due Date: 4/27/26 2:21 PM

Help Topic: Business Critical Outage

Last Message: 4/27/26 1:21 PM

Last Response: 4/27/26 1:28 PM

Ticket Thread (2)

Tasks

James Carter posted 4/27/26 1:21 PM

Good Afternoon,

No one in the office is able to send or receive emails as of 1:00 PM. This is affecting all departments and we have a long list of client communications pending, growing by the hour. This needs to be resolved immediately.

Created by James Carter 4/27/26 1:21 PM

Bryan Sykes assigned this to Lindsey Matthews 4/27/26 1:25 PM

Bryan Sykes posted 4/27/26 1:28 PM

Good Afternoon James,

This is Lindsey Matthews from the IT Support Tier II Team. I apologize for the inconvenience this Business Outage has caused for the office. I have received your ticket and will be looking into this issue as soon as possible.

Ticket is assigned to Lindsey Matthews

Validation: Critical outage immediately escalated to Emergency priority. The SLA engine calculated a 1-hour due date confirming SEV-A enforcement. Routed to IT Support Tier 2 and assigned to Lindsey Matthews from Systems Administration. Full audit trail logged with timestamps. Lindsey responded acknowledging business impact and committing to immediate investigation.

Section 6: Project Summary

What Was Built

- Fully deployed osTicket instance on a local XAMPP stack (Apache + PHP + MySQL)
- 4 departments mirroring a real IT organizational structure
- 3 cross-department teams for routing and escalation
- 4 agents across Tier 1, Tier 2, Network Operations, and Security
- 4 end users representing a simulated employee base
- 3-tier SLA system with automated due date calculation
- 6 help topics mapped to departments, SLA plans, and priority levels
- 4 automation filters for intelligent ticket routing without manual triage
- 4 end-to-end ticket workflows across all severity levels with full documentation

Skills Demonstrated

- Web application deployment on a local LAMP-equivalent stack
- Database administration via phpMyAdmin and MySQL CLI
- Helpdesk system architecture, configuration, and administration
- SLA design, enforcement, and automated due date calculation
- Ticket routing automation and filter logic
- End-to-end ticket lifecycle management (open → assigned → in-progress → resolved)
- Systematic troubleshooting and root cause analysis across four distinct issues
- Technical documentation and IT portfolio development

Portfolio Note: This lab was built entirely from scratch including independent troubleshooting of four distinct installation issues. All configuration decisions reflect real-world helpdesk best practices and are documented with evidence of work product throughout.